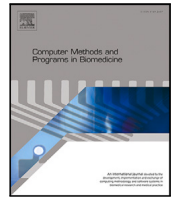




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Cross-Fusion Adaptive Feature Enhancement Transformer: Efficient high-frequency integration and sparse attention enhancement for brain MRI super-resolution[☆]

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ABSTRACT

Background and Objectives: High-resolution magnetic resonance imaging (MRI) is essential for diagnosing and treating brain diseases. Transformer-based approaches demonstrate strong potential in MRI super-resolution by capturing long-range dependencies effectively. However, existing Transformer-based super-resolution methods face several challenges: (1) they primarily focus on low-frequency information, neglecting the utilization of high-frequency information; (2) they lack effective mechanisms to integrate both low-frequency and high-frequency information; (3) they struggle to effectively eliminate redundant information during the reconstruction process. To address these issues, we propose the Cross-fusion Adaptive Feature Enhancement Transformer (CAFET).

Methods: Our model maximizes the potential of both CNNs and Transformers. It consists of four key blocks: a high-frequency enhancement block for extracting high-frequency information; a hybrid attention block for capturing global information and local fitting, which includes channel attention and shifted rectangular window attention; a large-window fusion attention block for integrating local high-frequency features and global low-frequency features; and an adaptive sparse overlapping attention block for dynamically retaining key information and enhancing the aggregation of cross-window features.

Results: Extensive experiments validate the effectiveness of the proposed method. On the BraTS and IXI datasets, with an upsampling factor of $\times 2$, the proposed method achieves a maximum PSNR improvement of 2.4 dB and 1.3 dB compared to state-of-the-art methods, along with an SSIM improvement of up to 0.16% and 1.42%. Similarly, at an upsampling factor of $\times 4$, the proposed method achieves a maximum PSNR improvement of 1.04 dB and 0.3 dB over the current leading methods, along with an SSIM improvement of up to 0.25% and 1.66%.

Conclusions: Our method is capable of reconstructing high-quality super-resolution brain MRI images, demonstrating significant clinical potential.

1. Introduction

High-resolution (HR) brain magnetic resonance imaging (MRI) is essential for disease diagnosis and therapy. [1–3]. However, inherent limitations in the MRI imaging mechanism often result in scattering and noise. Additionally, prolonged scanning times and patient positioning

constraints can lead to motion artifacts and distortions in the images [4, 5]. To overcome these challenges, super-resolution (SR) techniques have emerged as a prominent area of research. SR techniques aim to enhance the resolution of images [6–9], effectively compensating for deficiencies in the imaging process.

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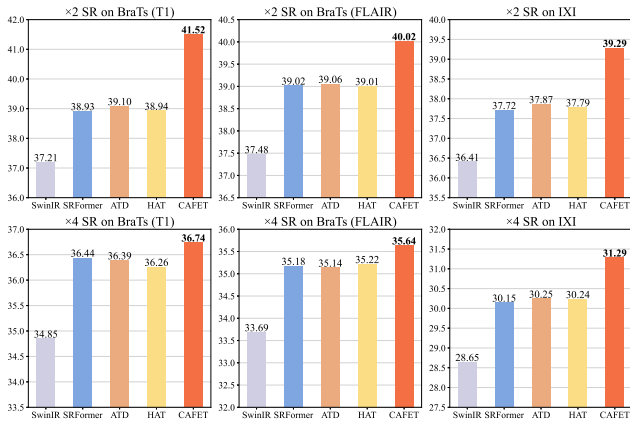


Fig. 1. Our method, CAFET, outperforms state-of-the-art methods such as SwinIR [22], SRFormer [23], ATD [24], and HAT [25], with a 0.3–2.4 dB PSNR improvement, highlighting its superiority.

Traditional SR reconstruction methods predominantly utilize interpolation and learning-based techniques. However, these methods lack adaptability to varying scaling factors, viewpoint changes, and complex lighting conditions, rendering them inadequate for practical applications [10,11]. Recently, deep learning-based single-image SR methods for natural images have shown remarkable progress in addressing the limitations of traditional approaches. Motivated by these advancements, deep learning-based SR reconstruction methods for medical MRI images have also emerged [6,12–17]. Some networks enhance the performance of convolutional neural networks (CNN) by introducing residual and dense connections [14,18,19]. Others incorporate attention mechanisms to capture long-range dependencies and contextual relationships, thereby further enhancing model performance [15,17,20–22]. These advanced methods have substantially enhanced MRI SR reconstruction results, providing more promising solutions for real-world clinical applications.

While attention mechanisms excel in MRI SR, particularly in capturing long-range dependencies, existing studies indicate a tendency to focus predominantly on low-frequency information [26–28]. Most existing studies focus on designing Transformer modules but fail to effectively utilize high-frequency information. For instance, Chen et al. [25] proposed utilizing mixed attention to capture additional features for attention computation, however, they did not address the difficulty that attention mechanisms have in concentrating on local information, leading to the loss of edge features during the reconstruction process. Furthermore, some methods [23,29] integrate self-attention with spatial-channel attention to improve feature representation capabilities; however, both self-attention and spatial-channel attention possess limitations in capturing texture details and edge information. Although some studies have attempted to integrate high-frequency and low-frequency information or introduce prior knowledge, these methods lack efficient fusion strategies. For example, Li et al. [26] employed a self-attention mechanism to fuse high-frequency and low-frequency information; however, when handling large-scale medical images or complex features, global self-attention can result in significant computational and memory overhead, as well as introduce noise and irrelevant information, thereby diminishing the model's robustness. Meanwhile, Zhang et al. [24] introduced a labeled dictionary to extract prior information from the data and adjust images through adaptive optimization steps, refining and enhancing the input features. However, the introduction of a labeled dictionary increases computational burdens, limits the model's flexibility, and decreases its resilience to interference. These limitations result in problems such as texture distortion, blurriness in details, and uneven reconstruction outcomes. Moreover, some networks employ window-based attention mechanisms

to capture long-range dependencies more efficiently [22,26,30], but these approaches lack direct interactions between adjacent windows. To overcome this limitation, several studies have introduced overlapping window attention mechanisms to improve interactions between neighboring windows [25,31]. However, in dense attention computation, overlapping windows not only process regions with insufficient information but also introduce redundant or irrelevant features, which cause noise interference and degrade performance.

To address these challenges, we propose a Cross-fusion Adaptive Feature Enhancement Transformer (CAFET) network. First, we design a Hybrid Fusion Block (HFB), which consists of a Hybrid Attention Block (HAB), a High-Frequency Enhancement Block (HFEB), and a Large-Window Fusion Attention Block (LWFAB). The HAB comprises a Shifted Rectangular Window Attention Block (SRWAB) and a Channel Attention Block (CAB) to effectively capture global information and enhance local fitting capabilities. The HFEB enhances high-frequency details, whereas the LWFAB expands the receptive field, captures more features, and efficiently integrates and refines those extracted from both the HFEB and the HAB, while simultaneously reducing computational overhead.

Then, we propose an Adaptive Sparse Overlapping Attention Block (ASOAB), which aims to enhance direct interaction between different windows and dynamically retain key information while filtering out redundant features. This block computes dense and sparse overlapping window attention through two branches and adaptively fuses them. Compared to traditional self-attention mechanisms, this design strengthens feature correlations, ensures attention sparsity, effectively eliminates redundant information, and enhances overall network performance.

As shown in Fig. 1, our method significantly outperforms the best existing methods on the PSNR metric, with improvements ranging from 0.3 dB to 2.4 dB, demonstrating high application potential. Our main contributions can be summarized as follows:

- We designed a framework that integrates the complementary features of CNN and Transformer, and devised a HAB and a HFEB to capture global and local information respectively. This design significantly enhances the performance of MRI SR tasks.
- The LWFAB is proposed to efficiently integrate feature information from both the HAB and the HFEB. This block not only expands the receptive field while significantly reducing computational overhead but also captures more features for participation in attention computation, ensuring that the reconstructed MRI images are distortion-free.
- The ASOAB is proposed to enhance direct interactions between features across windows. It adaptively strengthens feature relationships among tokens, dynamically preserves key information, and effectively eliminates redundant features.
- Experimental results on the BraTs and IXI datasets show that our method surpasses state-of-the-art techniques in MRI image reconstruction across different anatomical structures.

2. Related works

2.1. Medical image SR based on CNN

Inspired by the success of numerous CNN-based SR algorithms for natural images, an increasing number of SR methods have been developed for medical imaging. Qiu et al. [12] were the first to apply CNNs to the SR reconstruction of knee MRI, pioneering research in this field. Subsequently, Ren et al. [13] proposed a residual structure to improve feature extraction capabilities. Bao et al. [19] proposed applying error correction units to residual networks and utilizing dense connections to enhance the recursive residual network, thereby significantly improving reconstruction performance. Liu et al. [32] proposed a method to learn more discriminative feature representations using multiple

feature distillation connections. Additionally, they introduced shallow residual blocks to aid the network in acquiring more feature information. Gao et al. [6] combined residual and multi-channel connections to develop a Residual Connection Block that effectively integrates local features at multiple depths and enhances multi-level feature extraction. Sun et al. [33] designed a deep convolutional network that utilizes large kernel convolutions and incorporates two projection layers based on channel partitioning and reshuffling to effectively mix features. Additionally, they introduced Fused-MBConv in the proposed network to model various local features. To fully utilize the information contained in the input data, Li et al. [34] introduced a global feedback loop framework, CMISR, which combines implicit low-resolution (LR) units with advanced explicit SR units. By propagating error terms from the front to the back of the network, this framework significantly improves overall network performance. To reconstruct images with sharp edges, Liu et al. [35] proposed a cross-convolution approach that builds on traditional edge detectors to locate and represent edge features. They also designed a cross-convolution block that integrates feature normalization and channel attention mechanisms to learn the intrinsic correlations among features, demonstrating exceptional performance.

Although CNN-based methods have shown significant promise, medical images often feature intricate anatomical structures and non-local characteristics, such as long-range symmetry and interrelationships, because CNNs mainly depend on local feature extraction and lack the ability to model non-local information, this limitation can lead to a deficiency in the model's feature representation capability. Furthermore, these methods struggle to maintain the accuracy of global structures and preserve anatomical consistency. During feature extraction, these methods are prone to interference from redundant information, making it challenging to eliminate redundancy while retaining essential details.

2.2. Medical image SR based on Transformer

Unlike traditional CNNs, Transformers [36] capture long-range contextual dependencies through self-attention mechanisms, achieving significant advancements in the field of MRI reconstruction. Considering the interdependencies between channels, Zhang et al. [20] proposed a channel attention mechanism that adaptively reweights channel features, effectively enhancing reconstruction performance. Feng et al. [37] proposed a dual-branch multi-task network that combines reconstruction and SR tasks, and designed a task Transformer module to integrate the correlations between these tasks. However, the network encounters challenges in balancing the weights assigned to the different tasks, which ultimately leads to suboptimal reconstruction performance. Liu et al. [38] proposed the Swin Transformer, which employs a hierarchical architecture and incorporates a shifted window mechanism. This design restricts self-attention to local windows and enables cross-window connections, significantly enhancing computational efficiency. Additionally, Wang et al. [39] proposed a universal self-attention paradigm based on the principle of dense interactions, capable of simultaneously modeling pixel interactions in both spatial and channel dimensions. They also introduced a multi-scale interaction scheme to enhance effective interactions at both local and global scales. However, the network is highly sensitive to the quality of input images, and its performance may suffer significantly when the input image quality is low. To address this issue, Chen et al. [25] proposed enhancing the model's global statistics and local fitting capabilities using a hybrid attention mechanism, alongside a window-based cross-attention module to aggregate cross-window information. Although this network can capture more information for attention calculations, it continues to face challenges in effectively modeling edge textures. Wang et al. [17] proposed a dual-branch attention block that combines the long-range dependency modeling capability of Transformers with the ability of CNNs to capture high-frequency local information through channel

splitting. This approach is used to reconstruct undersampled images and synthesize missing sequences.

Although these methods address the effective utilization of global and local features in MRI reconstruction to some extent, they remain insufficient in extracting and utilizing high-frequency features and struggle to efficiently fuse global and local information. Additionally, these methods retain a significant amount of redundant features during reconstruction, leading to MRI images with insufficient detail and edge clarity. To overcome these limitations, our proposed method fully integrates the strengths of CNNs and Transformers to extract both high-frequency and low-frequency information. It adaptively retains critical features, enabling more efficient exploration of image information and significantly enhancing the detail and edge quality of MRI SR images.

2.3. Medical image SR with hybrid architecture

Recently, MRI SR methods that integrate CNNs and Transformers have emerged, offering strong support for reconstructing SR images characterized by clear textures and global consistency. Sun et al. [21] designed an attention-serial-fusion residual channel (RPC) structure that adaptively focuses on critical regions in MRI images and captures high-frequency information through residual learning. Xu et al. [15] proposed the use of the SADNet CNN to assist the Swin Transformer in reconstructing dynamic MRI and designed a novel reconstruction head to more effectively restore pixel intensities. Gao et al. [40] proposed a multi-branch CNN network focused on local feature extraction, which incorporates channel attention and spatial triggering mechanisms to reweight feature information from different branches for feature fusion and refinement. However, they did not consider the integration of global information, leading to challenges in maintaining structural consistency. In contrast to Gao et al. Li et al. [26] proposed the use of convolutional structures to extract high-frequency information while employing a Transformer module to capture low-frequency information, utilizing self-attention calculations to fuse global and local information. However, the global self-attention calculations during this fusion process significantly increase computational and memory overhead, as well as introducing noise and irrelevant information. Chen et al. [29] proposed alternately applying spatial and channel self-attention to capture global context and facilitate the aggregation of various local features. Additionally, they introduced adaptive interaction modules and spatial gate feedforward networks to enhance local feature aggregation. Zhang et al. [24] introduced a labeled dictionary to extract prior information from the data, adjusting images through adaptive optimization and employing a category-based self-attention mechanism to improve input features. However, the introduction of the dictionary increases computational burden, limits the model's flexibility, and diminishes its robustness to interference.

Although these methods incorporate high-frequency and low-frequency information, they lack effective feature fusion mechanisms, which makes it challenging to reconstruct fine anatomical structures and leads to significant computational overhead. To address this issue, our proposed method effectively captures more high-frequency and low-frequency features for attention calculations, and it integrates and refines these features efficiently without incurring extra computational costs, all while expanding the receptive field.

3. Methods

3.1. Overall architecture

As shown in Fig. 2, the overall framework comprises three key components: shallow feature extraction, deep feature extraction, and image reconstruction. Specifically, the input LR image $I_{LR} \in \mathbb{R}^{H \times W \times C_{in}}$ first undergoes shallow feature extraction to obtain the shallow features $F_s \in \mathbb{R}^{H \times W \times C}$, where C_{in} denotes the number of input channels, and C represents the number of intermediate feature channels. Next, the

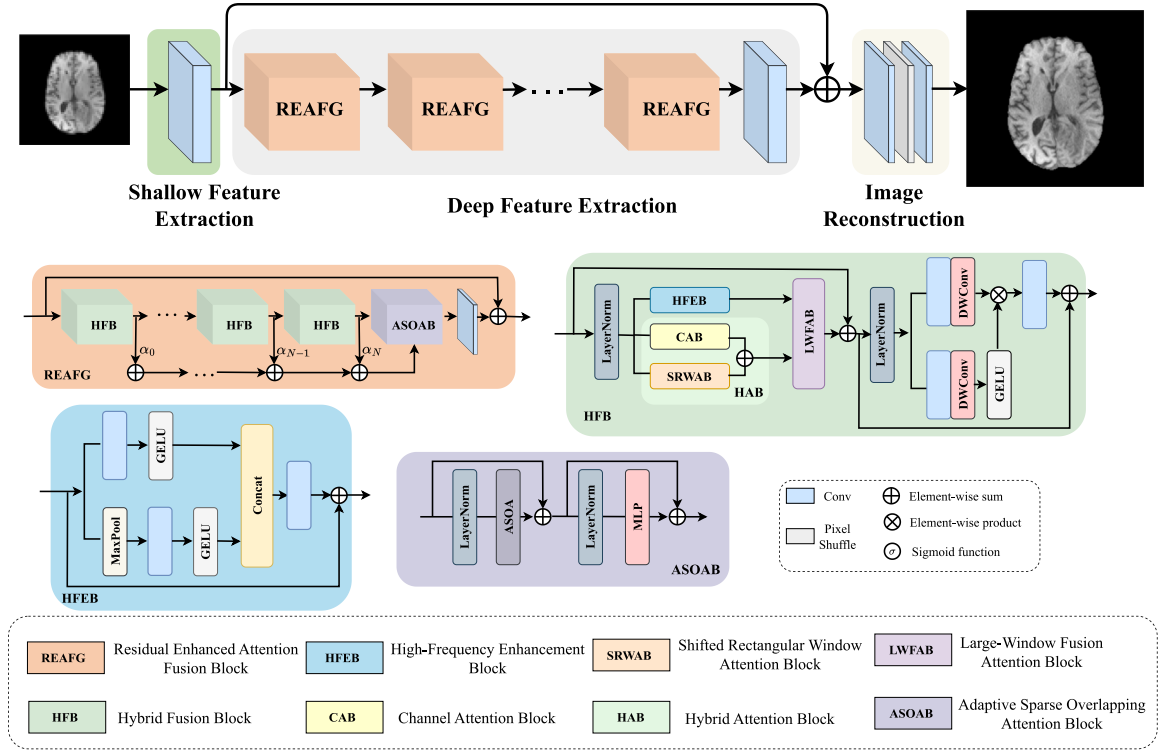


Fig. 2. The overall architecture of CAFET, where REAFB consists of HFB and ASOAB.

shallow features are processed through N Residual Enhanced Attention Fusion Blocks (REAFB) and a 3×3 convolutional layer to extract deep features F_d . Subsequently, the shallow features F_s and the deep features F_d are fused using a global residual connection, and the final HR image is generated through the image reconstruction module.

The REAFB consists of a HFB, an ASOAB, and a 3×3 convolutional layer with a global residual connection. The image reconstruction module employs the pixel shuffle [41] method to upsample the fused features, producing the SR image. The network parameters are optimized by training the model with the $\mathcal{L}_1$ loss function.

3.2. Hybrid Fusion Block

The goal of the HFB is to thoroughly extract and integrate high-frequency and low-frequency information within the image, thereby enhancing SR reconstruction. As illustrated in Fig. 2, the HFB primarily consists of a HFEF, a HAB, and a LWFAB.

3.2.1. High-Frequency Enhancement Block

The HFEF is designed to effectively capture edge detail information and enhance high-frequency information, as shown in Fig. 2. After processing through the initial LayerNorm layer, the input features generate the intermediate feature F_{in} . This feature F_{in} is subsequently split along the channel dimension and fed into two branches for high-frequency feature extraction and enhancement:

$$F_M, F_N = \text{Split}(F_{in}), \quad (1)$$

where $F_M, F_N \in \mathbb{R}^{H \times W \times C/2}$ represent the intermediate features. For F_M , we apply a 3×3 convolution and GELU activation to extract the local feature F_m , which can be expressed as:

$$F_m = \text{GELU}(\text{Conv}^{3 \times 3}(F_M)). \quad (2)$$

For F_N , we first apply max-pooling to capture the high-frequency components of the feature, followed by a 1×1 convolution and GELU

activation to enhance the high-frequency information. This process can be expressed as:

$$F_n = \text{GELU}(\text{Conv}^{1 \times 1}(\Phi(F_N))), \quad (3)$$

where Φ denotes max-pooling.

Finally, the two branches are concatenated along the channel dimension and fused using a 1×1 convolution. To minimize information loss, we introduce a residual connection. The entire process can be formulated as:

$$F_H = \text{Conv}^{1 \times 1}(\text{Concat}(F_m, F_n)) + F_{in}. \quad (4)$$

By extracting local features and enhancing high-frequency information, subtle structures within the image can be more effectively captured, which improves local contrast and facilitates the reconstruction of edges and textures, ultimately enhancing the discernibility of the image.

3.2.2. Hybrid Attention Block

The HAB employs shifted rectangular window attention and channel attention to capture global information and enhance representation capacity, as shown in Fig. 2.

In the anatomical MRI of the brain, there is a clear presence of global similarity features. To efficiently model the long-range dependencies in MRI, inspired by recent advancements [24–26], we introduce the SRWAB. As shown in Fig. 3, the SRWAB comprises the shifted rectangular window attention mechanism and MLPs. To more effectively capture information from different regions, the SRWAB is divided into Vertical Window Attention (referred to as HSAB when the window height r^h exceeds the width r^w) and Horizontal Window Attention (referred to as WSAB when the window width r^w exceeds the height r^h).

In the concrete implementation, we utilize multiple attention heads and assign HSAB and WSAB to the same number of attention heads in parallel for independent computation. Subsequently, the outputs are concatenated along the channel dimension to integrate information

from different windows. By expanding the area of attention computation, SRWAB can effectively capture pixel relationships in both horizontal and vertical directions within the brain, aggregating long-distance features and enhancing interactions between windows without increasing computational complexity. This ensures that the network can extract rich features across different upsampling scales, thereby enhancing the model's ability to understand context.

Given an input F_{in} , it is initially divided into $\frac{H \times W}{p_h \times p_w}$ non-overlapping rectangular windows. For each rectangular window feature F_{in}^i is projected into F_Q , F_K , and F_V , representing the query, key, and value vectors, respectively. The process is expressed as:

$$F_Q, F_K, F_V = F_{in}^i \mathbf{L}_Q, F_{in}^i \mathbf{L}_K, F_{in}^i \mathbf{L}_V, \quad (5)$$

where $\mathbf{L}_Q$, $\mathbf{L}_K$, and $\mathbf{L}_V$ are the projection matrices for the query, key, and value, respectively. Then, rectangular window attention is computed as:

$$Att(F_Q, F_K, F_V) = Softmax\left(\frac{F_Q F_K^T}{\sqrt{d}} + B\right) F_V, \quad (6)$$

where d denotes the channel dimension, and B represents the dynamic positional encoding. Attention operations are applied to all windows, and the final feature X_{att} is obtained through the MLP layer. It is important to note that the operations of WSAB and HSAB are identical.

Moreover, research [25,29] indicates that the channel attention mechanism effectively captures global information. Therefore, we embed a CAB in the model to enhance the representation capacity of visual features. To prevent potential conflicts between the CAB and SRWAB during network optimization, we multiply the output of the CAB by a small scaling factor α , ensuring the stability and consistency of the model. Finally, the scaled channel attention output is added to the output of the SRWAB to obtain the final output feature of the HAB F_{att} . The resulting operation can be expressed as:

$$F_{att} = \alpha CAB + X_{att}. \quad (7)$$

3.2.3. Large-Window Fusion Attention Block

The LWFAB, shown in Fig. 4, was designed to preserve edges and textures while integrating low-frequency and high-frequency information. This block enhances the model's ability to learn complex image structures and feature relationships, while simultaneously expanding the receptive field and lowering computational costs. Specifically, we divide the output features F_H from the HFEB and F_{att} from the HAB into M non-overlapping windows, denoted as $F'_H \in \mathbb{R}^{M \times L^2 \times C}$ and $F'_{att} \in \mathbb{R}^{M \times L^2 \times C}$, where L represents the side length of each window. Next, the linear layer encodes F'_H and F'_{att} , generating two query vectors $Q_H \in \mathbb{R}^{M \times L^2 \times C}$ and $Q_{att} \in \mathbb{R}^{M \times L^2 \times C}$. These two query vectors are then initially fused to form the final query vector Q . Simultaneously, the output F_{att} from the HAB is divided into windows in the same manner, and then the feature is projected and the number of channels is compressed to obtain the key matrix $K \in \mathbb{R}^{M \times L^2 \times C/r^2}$ and the value matrix $V \in \mathbb{R}^{M \times L^2 \times C/r^2}$, where r is the token reduction factor. The K and V matrices are then rearranged to generate the permuted key matrix $K' \in \mathbb{R}^{(M \times L^2)/r^2 \times C}$ and the permuted value matrix $V' \in \mathbb{R}^{(M \times L^2)/r^2 \times C}$. In this way, the reduction factor r effectively reduces the computational cost of large window attention while keeping the channel dimension C unchanged, ensuring that each attention head has sufficient expressive power when generating the attention map. Finally, attention is computed to fully integrate global and local information:

$$F_{out} = Softmax\left(\frac{Q K_T'}{\sqrt{d_k}} + B\right) V', \quad (8)$$

where B denotes the relative position encoding, K_T' is the transpose of K' , and d_k represents the channel dimension. By reducing the channel dimensions of the K and V matrices and mapping a portion

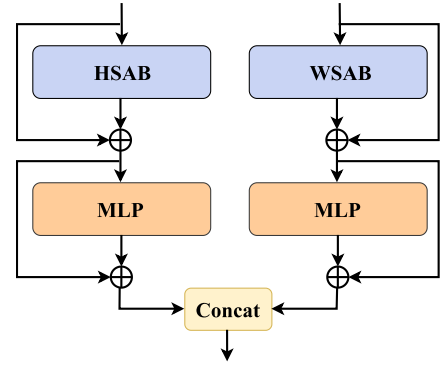


Fig. 3. The structure of SRWAB is illustrated in the figure, consisting of two rectangular window attention blocks and two MLPs. HSAB and WSAB are implemented in parallel across different attention heads.

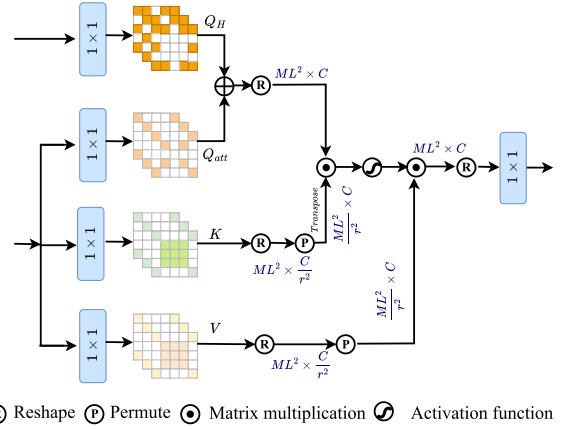


Fig. 4. The schematic diagram of the LWFAB illustrates that the input features are composed of the output F_{att} from the HAB and the output F_H from the HFEB. In the attention computation, Q_H and Q_{att} are generated from the projections of F_H and F_{att} , respectively, while K and V are derived from the projections of F_{att} .

of the spatial information to the channel dimension through a rearrangement operation, the computational complexity of large-window attention is significantly reduced. This process also captures more information, enabling a deep fusion of high-frequency and low-frequency information.

Next, the fused feature F_{out} is passed into a feed-forward network for full feature aggregation. As shown in Fig. 2, a 1×1 convolution is employed to aggregate channel-wise information, while a 3×3 convolution is used for spatial information aggregation. Additionally, a gating mechanism enhances information encoding, resulting in the fully fused features F'_{out} . The process can be expressed as:

$$F'_{out} = Conv^{1 \times 1} [GELU(\Omega(F_{out})) \otimes \Omega(F_{out})] + F_{out}, \quad (9)$$

where Ω represents convolution and depthwise convolution operations, while $\otimes$ denotes element-wise multiplication.

This design effectively refines the global information obtained from HAB while deeply integrating global and local features. By mapping partial spatial information to the channel dimension, this design effectively expands the receptive field without increasing computational load, capturing a larger number of features for attention computation facilitates the establishment of more intricate relationships between features, thereby enhancing the model's ability to represent them effectively.

3.3. Adaptive Sparse Overlapping Attention Block

To facilitate direct interactions between windows and reduce the introduction of irrelevant features, we propose the ASOAB. As illustrated in Fig. 2, ASOAB comprises an adaptive sparse overlapping attention (ASOA) and a MLP. To adaptively enhance features while retaining key information, we multiply the output of each HFB by a learnable residual weighting factor, forming a learnable residual branch that produces a residual output. The residual output is subsequently added to the output of the N th HFB through a residual connection, producing the final feature representation, which is then used as the input to the ASOAB. This process can be formulated as:

$$X = \alpha_0 F_{out}^{0'} + \alpha_1 F_{out}^{1'} + \dots + \alpha_N F_{out}^{N'} + F_{out}^{N'}, \quad (10)$$

where X denotes the input to the ASOAB, $[\alpha_0, \alpha_1, \dots, \alpha_N]$ represents a set of learnable weighting factors, and F_{out}^i ($i = 0, 1, 2, \dots, N$) denotes the output of each HFB.

For the ASOA, the structure is shown in Fig. 5. It is implemented by expanding input features into blocks of different window sizes and performing shifted window attention calculations with kernel size M_0 and stride size M on these blocks. The input features are projected into query (Q), key (K), and value (V) vectors. The Q vector is divided into $\frac{HW}{R^2}$ non-overlapping windows of size $R \times R$, whereas the K and V vectors are divided into $\frac{HW}{R^2}$ overlapping windows of size $R_0 \times R_0$. To ensure consistency in window size, zero-padding of $\frac{\gamma R}{2}$ is applied to the K and V vectors. The calculation for R_0 is as follows:

$$R_0 = (1 + \gamma)R. \quad (11)$$

Subsequently, attention is computed, yielding the dense attention feature Att_d , as shown in Eq. (6).

Additionally, traditional attention mechanisms typically employ the softmax function for activation. However, softmax normalization retains all similarity scores between the query and key tokens, despite the fact that not all query tokens are strongly related to their corresponding key tokens. This operation may introduce irrelevant noise interactions, which can negatively impact the network's performance. To address this issue, we propose a RELU-based attention mechanism designed to filter out features with low matching scores between the query and key tokens. Moreover, we square the RELU output to enhance positive correlations and effectively reduce the interference of irrelevant information, while ensuring the sparsity of the attention mechanism. The attention computation can be expressed as:

$$Att_s = RELU^2 \left(\frac{QK^T}{\sqrt{d}} + B \right) V. \quad (12)$$

Given that the RELU-based self-attention mechanism may lead to excessive sparsity [42,43], potentially causing the loss of useful information, we retain the original softmax activation operation and propose a dual-branch attention to balance sparsity and density in the attention mechanism. It should be noted that in [43], a softmax-like operation is employed to compute the weight factors, thereby imposing constraints on both branches. This may limit the overall performance of the model when one branch performs poorly, even if the other branch excels, consequently resulting in diminished overall performance. Additionally, dynamic weight adjustments between the branches can lead to model instability, as the learning rates and feature aggregation effects may differ, ultimately reducing convergence speed during training. In contrast to [43], we impose constraints solely on the sparse branch, facilitating the adaptive fusion of the two branches and enabling the network to dynamically adjust while retaining essential information. The final attention result X_{final} can be expressed as:

$$X_{final} = Att_d + W \cdot Att_s, \quad (13)$$

where W represents an adaptive weight that balances the contributions of the sparse and dense attention branches.

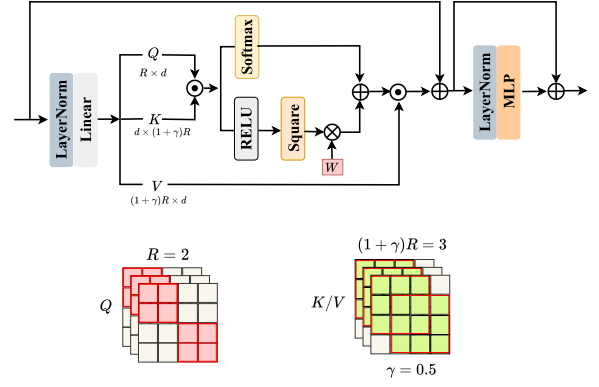


Fig. 5. The proposed ASOA is composed of a dense attention branch utilizing the Softmax activation function and a sparse attention branch utilizing the RELU activation function. In this framework, W represents the learnable weight factor, R denotes the window size, γ refers to the scaling factor, and d indicates the channel dimension.

This design effectively reduces interference from redundant information resulting from dense window attention computation during the super-resolution reconstruction process while facilitating direct interactions between different windows, enabling the model to learn diverse feature combinations. Furthermore, it maximally preserves essential features, minimizes the introduction of noise, and dynamically adjusts the sparsity of relevant input tokens.

4. Results

4.1. Datasets

To train and evaluate the proposed network and verify its SR performance on different MRI-weighted images, we randomly selected 12,230 T1-weighted image slices and 9,629 FLAIR sequence image slices that contain pathological features from the BraTS2019 brain dataset [44], where all slices had a resolution of 240×240 . Additionally, to assess the model's reconstruction capability on T2-weighted images, we randomly selected 9232 slices from the T2-weighted images in the IXI dataset (<https://brain-development.org/ixi-dataset/>), each with a resolution of 256×256 . Both datasets were divided into training, validation, and testing subsets in a 7:2:1 ratio. To meet the requirements for network training, we applied bicubic downsampling on the HR images to generate paired datasets for $\times 2$ and $\times 4$ downsampling, which were used for the training and testing of the SR tasks.

4.2. Implementation details

The proposed method was implemented using PyTorch on a single NVIDIA RTX 3090 GPU with 24 GB of memory. The network was trained for 120 epochs with the $\mathcal{L}_1$ loss function and the Adam optimizer. The initial learning rate was set to 1×10^{-5} and gradually decayed during the training process. In the CAFET network architecture, the number of RHAFGs and HFBs was set to 6, and the number of channels was set to 180. For the hyperparameter settings in the proposed modules, similar to Chen et al. [25], the weighting factor α in the CAB of the HAB was set to 0.01. The window size in SRWAB was set to 4×16 and 16×4 , the number of attention heads was set to 6, the reduction factor r in LWFAB was set to 2, and the window size was set to 16. The overlapping ratio γ in ASOAB was set to 0.5. Additionally, PSNR and SSIM metrics were used to evaluate the performance of SR reconstruction.

Table 1

PSNR and SSIM are compared quantitatively under $\times 2$ and $\times 4$ upsampling on the BraTS and IXI datasets. The best results are marked in **red**, and the second-best in **blue**, against state-of-the-art methods.

Methods	BraTS (T1) [44]				BraTS (FLAIR) [44]				IXI			
	$\times 2$		$\times 4$		$\times 2$		$\times 4$		$\times 2$		$\times 4$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Bicubic	31.89	0.8645	30.32	0.8340	31.14	0.8544	29.58	0.8195	26.61	0.7699	25.15	0.6908
SRCNN [45]	36.76	0.9718	34.15	0.9531	35.97	0.9706	32.85	0.9337	30.79	0.9074	27.39	0.8201
FSRCNN [46]	35.93	0.9688	33.15	0.9441	35.43	0.9678	32.85	0.9337	30.21	0.8969	26.18	0.7712
RFDN [32]	37.36	0.9753	34.06	0.9521	37.48	0.9742	33.69	0.9274	31.59	0.9211	28.20	0.8461
Cross_srn [35]	37.35	0.9746	34.09	0.9513	36.69	0.9741	33.12	0.9354	31.71	0.9209	27.15	0.8312
$T^2 - Net$ [37]	37.26	0.974	34.11	0.9513	36.34	0.9722	33.09	0.9368	31.67	0.9171	27.99	0.8389
LBNet [40]	36.16	0.9685	32.93	0.9334	35.44	0.9669	32.35	0.9340	31.06	0.9053	27.71	0.8265
DITN [47]	37.10	0.9672	34.14	0.9513	36.31	0.9719	33.05	0.9401	31.24	0.9141	27.90	0.8360
VapSR [48]	37.12	0.974	34.07	0.9518	36.71	0.9741	33.36	0.9411	31.69	0.9221	28.03	0.8421
SwinIR [22]	37.21	0.9831	34.85	0.9607	37.48	0.9742	33.69	0.9274	36.41	0.9716	28.65	0.8793
SAFMN [49]	36.19	0.9695	33.50	0.9502	35.92	0.9706	32.72	0.9396	31.34	0.9155	27.84	0.8356
ShuffleMixer [33]	38.32	0.9843	35.91	0.9635	38.75	0.9766	35.05	0.9358	37.69	0.9750	29.91	0.8904
ESRT [50]	37.30	0.9746	34.07	0.9522	36.84	0.9741	33.19	0.9406	31.55	0.9199	27.92	0.8375
OmniSR [39]	37.57	0.9753	34.22	0.9533	36.81	0.9746	33.45	0.9416	31.77	0.9235	28.38	0.8506
DAT [29]	39.05	0.9854	36.36	0.9658	39.08	0.9775	35.21	0.9371	37.91	0.9760	30.27	0.8988
CRAFT [26]	37.46	0.9830	34.24	0.9546	37.23	0.9732	33.18	0.9213	37.52	0.9753	29.02	0.8853
SRFormer [23]	38.93	0.9853	36.44	0.9660	39.02	0.9774	35.18	0.9369	37.72	0.9756	30.15	0.8967
HAT [25]	38.94	0.9853	36.26	0.9658	39.01	0.9772	35.22	0.9374	37.79	0.9757	30.24	0.8979
LKFN [51]	38.45	0.9842	36.14	0.9645	38.73	0.9760	35.01	0.9355	37.44	0.9740	29.92	0.8910
ATD [24]	39.10	0.9855	36.39	0.9660	39.06	0.9774	35.14	0.9366	37.87	0.9760	30.25	0.8985
CAFET (Ours)	41.52	0.9871	36.74	0.9685	40.02	0.9787	35.64	0.9418	39.29	0.9802	31.29	0.9154

4.3. Experimental results

To validate the effectiveness of our method, we compared it with several CNN-based SR methods, including the original CNN method SRCNN [45]; the FSRCNN [46] method, which utilizes residual groups; the Cross_srn [35] method, which employs cross convolutions; the RFDN [32] method, based on residual distillation networks; the ShuffleMixer [33] method, designed with deep convolution and channel-splitting shuffling; and the LKFN [51] method, enhanced with large-kernel convolutions. Additionally, we compared several Transformer-based methods, including $T^2 - Net$ [37], which designs a multi-task transformer; DITN [47], which proposes lightweight self-attention computations; SwinIR [22], which employs a shifted window attention mechanism; SRFormer [23], which implements lightweight enhancements to the window attention mechanism; SAFMN [49], which introduces spatially adaptive feature modulation; DAT [29], which computes attention both between and within feature blocks for feature aggregation; and HAT [25], which employs mixed attention mechanisms to involve a larger number of pixels in the computations. Finally, we compared our method with hybrid SR approaches that integrate CNN and Transformer architectures to demonstrate its superiority. This includes LBNet [40], which utilizes symmetric CNNs to extract local information while integrating recursive transformers for global context; VapSR [48], which expands the attention computation range and substitutes deep convolutions with large kernel convolutions; ESRT [50], which dynamically adjusts feature map sizes with CNNs while capturing dependencies through transformers; CRAFT [26], which employs CNNs to obtain high-frequency priors while refining global features through attention; and ATD [24], which utilizes dictionaries to acquire priors and uses the attention mechanism to adaptively adjust feature weights.

In contrast to these methods, our approach effectively captures both high-frequency and low-frequency information while expanding the attention computation range to fully fuse features at various levels. Furthermore, it enhances direct interactions between various attention computation regions, enabling the model to learn diverse combinations of features while ensuring the sparsity of attention computations. The experimental results presented in Table 1 indicate that our method outperforms existing approaches across various datasets and modalities, thereby demonstrating its effectiveness.

4.4. Quantitative results

Table 1 provides a quantitative comparison between our proposed method and other SR methods under two upsampling factors ($\times 2$ and $\times 4$). The evaluations were performed on the BraTS tumor dataset and the IXI healthy brain dataset. Compared to existing methods, our approach demonstrates substantial improvements in both PSNR and SSIM metrics. Specifically, PSNR improvements range from 0.3 dB to 2.42 dB, while SSIM increased by a maximum of 0.0166. Across the same datasets and upsampling factors, our method consistently outperforms all compared methods. On the BraTS dataset for the T1 modality, our method achieves PSNR and SSIM values of 41.52 dB and 0.9871 for the $\times 2$ upsampling factor, and 36.74 dB and 0.9685 for the $\times 4$ upsampling factor. Notably, for the $\times 2$ upsampling factor, our method surpasses the state-of-the-art ATD method by 2.42 dB in PSNR and 0.0016 in SSIM. For the $\times 4$ upsampling factor, compared to the SRFormer method, our approach demonstrates improvements of 0.3 dB in PSNR and 0.0025 in SSIM. In the FLAIR pathological sequences of the BraTS dataset, our method achieved PSNR and SSIM values of 40.02 and 0.9787 at $\times 2$ upsampling, and 35.64 and 0.9418 at $\times 4$ upsampling, respectively. At $\times 2$ upsampling, the PSNR of our method improved by 0.94 dB compared to the state-of-the-art method DAT, while the SSIM increased by 0.0012. At $\times 4$ upsampling, our method enhanced the PSNR by 0.42 dB compared to the leading method HAT, the SSIM value exceeds that of the second-place method, OmniSR, by 0.0002.

On the IXI dataset for the T2 modality, our method achieves PSNR and SSIM values of 39.29 dB and 0.9802 for the $\times 2$ upsampling factor, and 31.29 dB and 0.9154 for the $\times 4$ upsampling factor. For the $\times 2$ upsampling case, our method surpasses the second-best performing method by 1.39 dB in PSNR and 0.0042 in SSIM. For the $\times 4$ upsampling factor, our method demonstrates an improvement of 1.04 dB in PSNR and 0.0166 in SSIM compared to the second-best method. The substantial performance improvements can be attributed to the effective integration of CNN and Transformer modules in our method, which fully leverages the strengths of both architectures. This approach effectively eliminates redundant information during the reconstruction process, thereby improving the model's ability to reconstruct high-quality images.

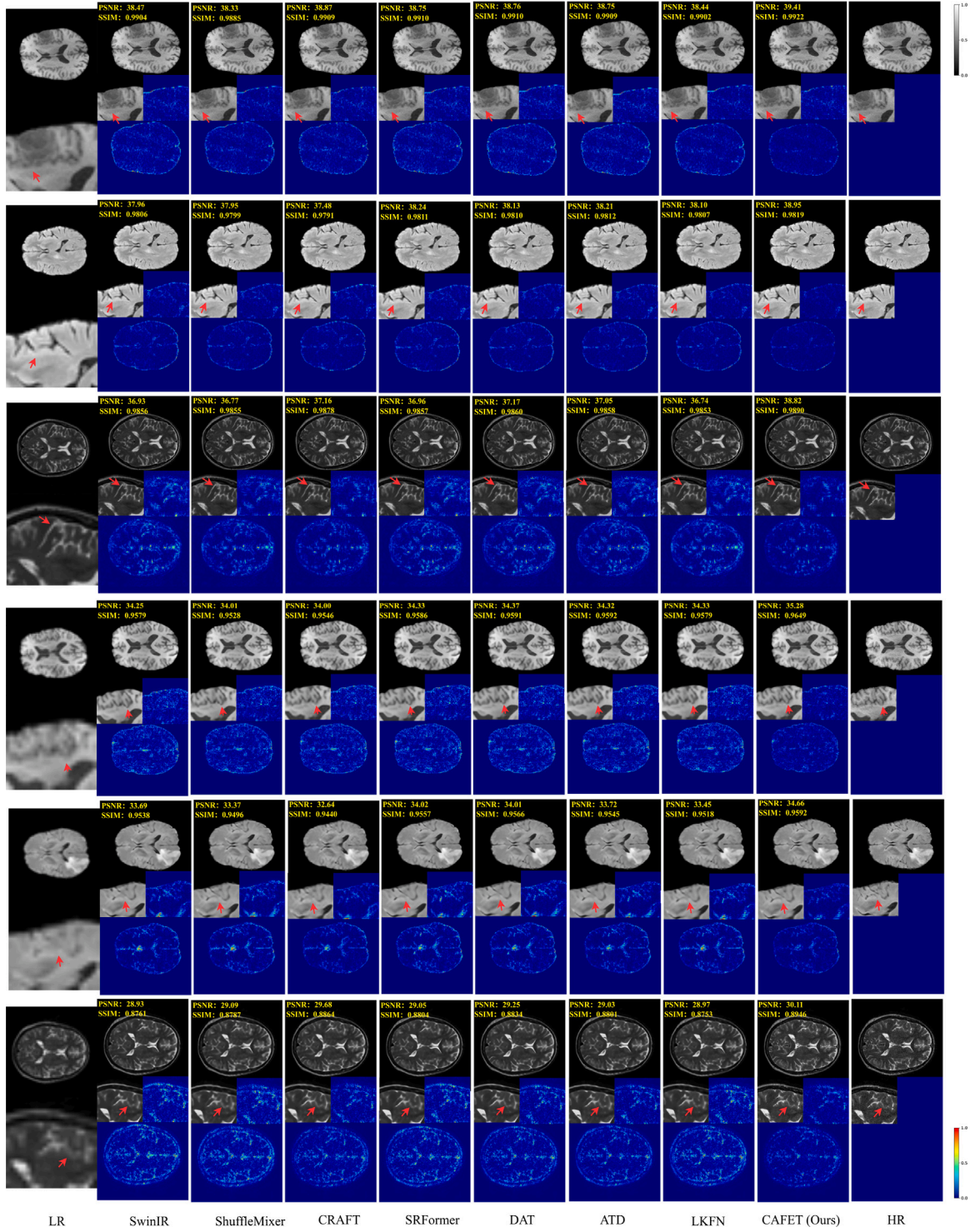


Fig. 6. A visual comparison was conducted under $\times 2$ and $\times 4$ upsampling factors. The first and second rows illustrate the upsampling reconstruction results for the pathological T1 and FLAIR modalities at a factor of $\times 2$, respectively. The third row presents the upsampling reconstruction results for the T2 modality at $\times 2$. The fourth and fifth rows illustrate the upsampling reconstruction results for the pathological T1 and FLAIR modalities at a factor of $\times 4$ respectively, while the sixth row presents the upsampling reconstruction results for the T2 modality at $\times 4$. PSNR/SSIM values are calculated on patches to reflect performance differences.

Finally, to validate the significant differences between our method and other approaches, we conducted a significance test on the experimental results. Specifically, we performed a Wilcoxon signed-rank test by comparing the list of PSNR quality metric values obtained from the reconstruction of the model's test set with the PSNR values from other state-of-the-art models. The results, as shown in Table 2, indicate that

all p-values are less than 0.05, demonstrating a significant difference between our method and the other approaches.

4.5. Qualitative results

Fig. 6 presents the SR reconstruction results achieved with $\times 2$ and $\times 4$ upsampling. The first and second rows display the $\times 2$ upsampling

Table 2

In the FLAIR pathological sequences of the BraTS dataset, we present a comparison of FLOPs, inference speed (Speed), and PSNR values at a $\times 4$ upsampling factor, along with the p-values obtained from the Wilcoxon signed-rank test. Bold text indicates the optimal results, while underlined text denotes the suboptimal results, with performance assessed based on PSNR values.

Methods	FLOPs	Speed	PSNR	p-value
SwinIR	47.12G	80.3 ms	33.69	0.000608
SRFormer	548.37G	96.57 ms	35.18	0.002369
CRATF	26.11G	142.7 ms	33.18	0.004642
DAT	549.56G	153.6 ms	35.21	0.009528
HAT	550.03G	648.7 ms	<u>35.22</u>	0.016218
ATD	748.91G	201.9 ms	35.14	0.000122
Ours	549.05G	158.6 ms	35.64	—

reconstruction results for the T1 and FLAIR pathological sequences, respectively. The third row presents the $\times 2$ upsampling reconstruction results for the T2 modality. The fourth and fifth rows illustrate the $\times 4$ upsampling reconstruction results for the T1 and FLAIR pathological sequences, while the sixth row presents the $\times 4$ upsampling reconstruction results for the T2 modality. To provide a more intuitive comparison of the reconstruction performance of various methods, we also include error maps. In these error maps, darker colors denote smaller reconstruction errors, signifying better recovery performance, while lighter colors indicate larger errors.

As shown in Fig. 6, for T1-weighted images, our method is able to maximally restore anatomical details in MRI under low upsampling factors while preserving the original information intact. In contrast, methods such as CRAFT, SRFormer, and DAT struggle to effectively reconstruct edge details. At higher upsampling factors, our method reconstructs most details successfully without introducing artifacts, whereas ATD and LKFN exhibit problems such as detail blurring and artifact generation. In the pathological FLAIR modality, our method effectively reconstructs the detailed textures surrounding the lesions at low upsampling factors, preventing excessive smoothing and accurately restoring the microstructures of normal regions. In contrast, methods such as LKFN and ShuffleMix demonstrate artifacts and distortions near the lesions. At high upsampling factors, our method successfully avoids the loss of high-frequency information and deformation of the lesions, while methods such as SwinIR and ATD encounter challenges related to excessive smoothing and the distortion of edge details.

For T2-weighted images, at low upsampling factors, our method significantly outperforms SwinIR and HAT in reconstructing global features and restoring complex shapes. With higher upsampling factors, our method maintains structural integrity and surpasses other approaches in handling details and boundary features. In comparison, methods such as SRFormer and ATD display issues with over-smoothing and boundary blurring. Across various imaging modalities and degradation levels, our model consistently exhibits superior performance, offering robust support for enhancing image quality and advancing medical image analysis.

4.6. Ablation study

4.6.1. Effectiveness of HFEB and HAB

To validate the contributions of the HFEB and HAB, we performed ablation experiments by individually removing these blocks to evaluate their impact on overall performance. As shown in Table 3, on the BraTS dataset with a $\times 4$ downsampling factor, the removal of HFEB and HAB led to PSNR decreases of 0.06 dB and 1.2 dB, respectively.

Additionally, we visualized the features learned by the HFEB and HAB at different scales, as presented in Fig. 7(b). Although the performance improvement of HFEB is relatively modest, the feature maps indicate that it effectively captures a substantial amount of texture and edge details, which are crucial for restoring image details and edge features. In contrast, HAB demonstrates a substantial improvement in

Table 3

Quantitative results of the ablation study on HFEB and HAB under $\times 4$ upsampling on the BraTS dataset. **Bold** indicates the optimal value.

HFEB	HAB	PSNR	SSIM
$\times$	$\checkmark$	36.71	0.9683
$\checkmark$	$\times$	36.65	0.9682
$\checkmark$	$\checkmark$	36.74	0.9685

Table 4

Ablation experiments on the inference speed and FLOPs of LWFAB under a $\times 4$ upsampling factor on the BraTS dataset, where $LWFAB_v$ indicates the removal of the rearrangement operation. **Bold** indicates the optimal value.

LWFAB	$LWFAB_v$	FLOPs	Speed
$\times$	$\checkmark$	92.72G	194.2 ms
$\checkmark$	$\times$	81.62G	158.6 ms

performance. As shown in Fig. 7(a), at the $\times 2$ and $\times 4$ upsampling factors, HAB effectively learns global information, maintains structural and spatial consistency, and demonstrates a strong ability to capture global features. To further evaluate the impact of each block, we generated error maps to visually assess their effects on image reconstruction. As illustrated in Fig. 8, the detailed views within the green boxes, along with the corresponding error maps, demonstrate that when either the HFEB or the HAB is used in isolation, the network is only able to learn local or global information, limiting its ability to achieve an optimal feature representation.

4.6.2. Effectiveness of LWFAB

To evaluate the effectiveness of the LWFAB, we performed ablation experiments by removing the LWFAB and performing feature fusion using element-wise addition. Specifically, we directly added the outputs of the HFEB and the HAB, and replaced the LWFAB with a 3×3 convolution operation. As shown in Table 5, removing the LWFAB caused a PSNR reduction of 0.29 dB on the BraTS dataset with $\times 4$ downsampling. The experimental results indicate that direct feature fusion through element-wise addition limits the network's ability to integrate high-frequency and low-frequency information effectively, leading to suboptimal feature representations. As shown in Fig. 7(c), we present the features learned by the LWFAB at different scales. The feature maps demonstrate that LWFAB possesses a large receptive field, effectively refining and integrating both global and local features, and is capable of extracting deeper-level information even with a $\times 4$ upscaling factor. This enables the model to capture the deep feature relationships between high-frequency and low-frequency information more effectively, thereby achieving effective fusion. By expanding the receptive field, the LWFAB effectively extracts more critical information and enables deeper fusion of high-frequency and low-frequency information. This enables the model to more effectively capture complex image structures and feature relationships.

Finally, to verify that the LWFAB significantly reduces computational overhead while expanding the receptive field, we conducted ablation experiments. Specifically, we removed the rearrangement operation and employed a standard window attention mechanism to process the key and query vectors, eliminating the token reduction factor r and not mapping some spatial information to the channel dimension. The experimental results are presented in Table 4, where we report the FLOPs and inference speed of the model. Compared to the ablation version, the LWFAB achieved a reduction of 9.1G in FLOPs and an improvement of 35.6 ms in inference speed. These results convincingly demonstrate the effectiveness of the LWFAB in reducing computational overhead and enhancing inference speed.

4.6.3. Effectiveness of ASOAB

We conducted ablation experiments to validate the effectiveness of the ASOAB. In these experiments, we directly removed the ASOAB

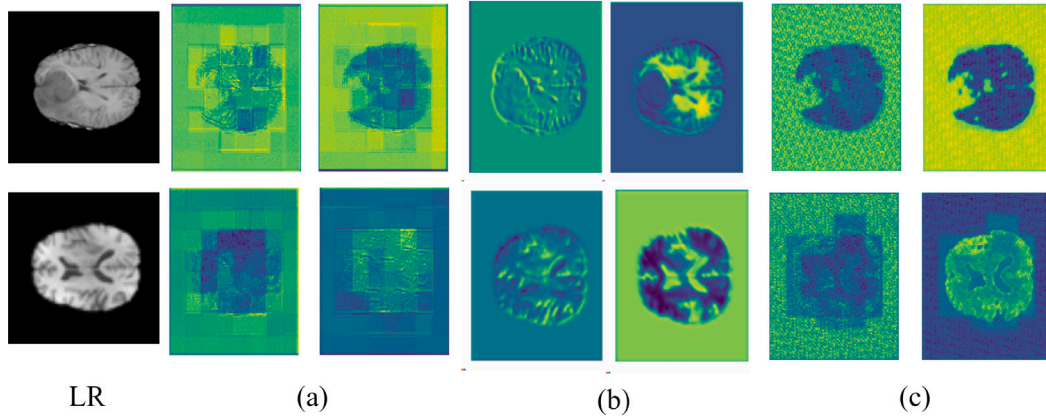


Fig. 7. In the feature visualization figures for the T1 pathological sequences of the BraTS dataset, LR represents the input image. The first row displays the feature maps at a 2× upsampling factor, while the second row presents the feature maps at a 4× upsampling factor. (a) Visualization of the output features from HAB. (b) Visualization of the output features from HFEB. (c) Visualization of the output features from LWFAB.

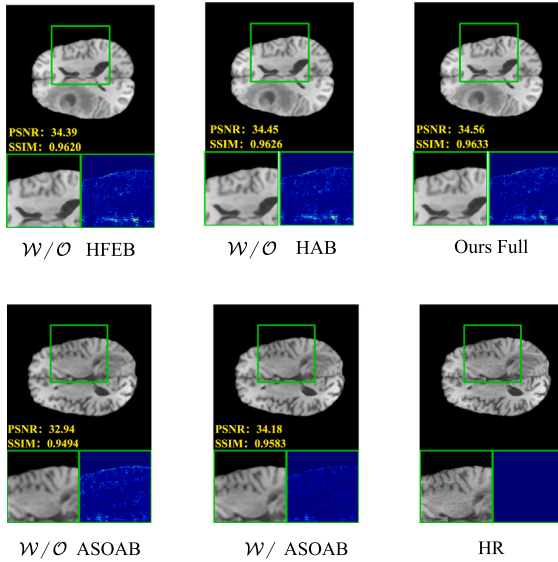


Fig. 8. Ablation visualization comparison of HFEB, HAB, and ASOAB.

Table 5

Quantitative results of ablation study on LWFAB and ASOAB with ×4 upsampling on the BraTS dataset, $ASOAB_{3 \times 3}$ denotes replacing ASOAB with a 3×3 convolution and performing feature fusion via element-wise addition. **Bold** indicates the optimal value.

LWFAB	$ASOAB_{3 \times 3}$	PSNR	SSIM
×	✓	36.45	0.9669
✓	×	35.41	0.9609
✓	✓	36.74	0.9685

for comparison. As shown in Table 5, removing the ASOAB resulted in a PSNR drop of 1.33 dB. Furthermore, as illustrated in Fig. 8, a visual analysis of the experimental results was performed. When the ASOAB was removed, the reconstructed images exhibited a significant decline in detail clarity compared to both the original images and the version with ASOAB, showing noticeable blurriness and artifacts. These findings demonstrate that ASOAB is crucial for eliminating redundant features, enhancing inter-window interactions, and dynamically preserving critical information. The experimental results strongly validate the effectiveness of the ASOAB in significantly enhancing image reconstruction quality.

Table 6

On the BraTS dataset, with a 4×upsampling factor, an ablation study was conducted on the dual activation attention branch in ASOA and the learnable residual weighting branch between the HFBs. Specifically, $CAFET_D$ refers to the dual activation function attention branch in ASOA, while $CAFET_W$ indicates the learnable residual weight branch between HFBs. **Bold** indicates the optimal value.

$CAFET_D$	$CAFET_W$	PSNR	SSIM
×	×	36.42	0.9609
✓	×	35.39	0.9608
×	✓	36.41	0.9669
✓	✓	36.74	0.9685

4.6.4. Effectiveness of learnable residual factors between multiple HFBs and dual activation functions in ASOAB

To validate the effectiveness of the dual activation function attention branch architecture in ASOAB and the learnable residual weighting branch across multiple HFBs for image reconstruction tasks, we performed an ablation study on both components. Four experimental configurations were designed to systematically evaluate the performance of the proposed method: (1) a single activation function attention architecture with a learnable residual weighting branch, (2) a single activation function attention architecture without a learnable residual weighting branch, (3) a dual activation function attention architecture with a learnable residual weighting branch, and (4) a dual activation function attention architecture without a learnable residual weighting branch. In the experiments, the single activation function utilized Softmax, whereas the dual activation function employed a combination of Softmax and RELU activations. The experiments were conducted on the BraTS dataset for ×4 downsampling image reconstruction tasks, with the results summarized in Table 6.

The experimental results show that the dual activation function attention architecture with a learnable residual weighting branch outperforms the single Softmax activation attention architecture with a learnable residual weighting branch, achieving a PSNR gain of 0.33 dB. To evaluate the effectiveness of dual activation function attention in eliminating redundant information, we plotted their entropy maps for both architectures, as illustrated in Fig. 9. The regions highlighted in the entropy maps indicate areas of redundant information. Clearly, the images reconstructed using the single activation function architecture display more redundant information, while those reconstructed with the dual activation function architecture show reduced redundancy. This confirms the effectiveness of the dual activation function attention branch in eliminating redundant information and improving reconstruction performance. Furthermore, the dual activation function attention architecture with a learnable residual weighting branch achieves a PSNR gain of 1.35 dB compared to its counterpart without a

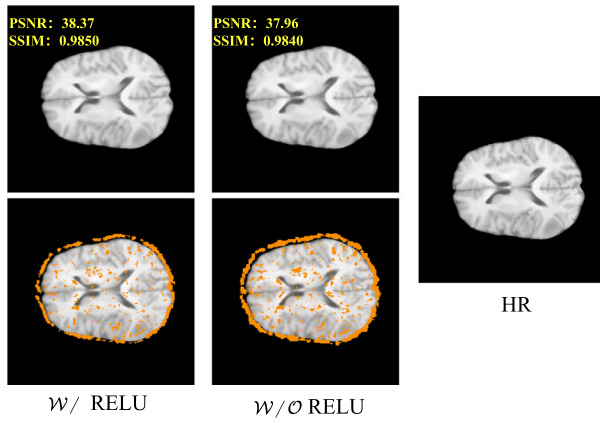


Fig. 9. Entropy maps of reconstructed images from single and dual activation function attention architectures with learnable residual weight branches, where $\mathcal{W} \setminus RELU$ denotes the reconstruction results from the dual activation function architecture, and $\mathcal{W} \setminus O RELU$ represents the reconstruction results from the single activation function architecture.

learnable residual weighting branch, further highlighting the necessity and effectiveness of incorporating this component to dynamically retain critical information. These results demonstrate that the combination of the dual activation function attention branch architecture and the learnable residual weighting branch proposed in this study significantly enhances the performance of MRI image reconstruction.

5. Discussion

SR techniques in MRI imaging significantly enhance image resolution and detail, thereby improving the performance of subsequent clinical tasks. In the field of image segmentation, SR images provide clearer representations of anatomical structures and pathological boundaries, enabling segmentation algorithms to identify tumors, organs, and other critical structures with greater accuracy. This enhancement reduces the likelihood of false positives and missed detections, offering clinicians more reliable diagnostic information [1,16]. Furthermore, in image classification tasks, SR enhances feature distinguishability, allowing classification models to achieve higher accuracy in identifying various diseases, such as brain tumors and neurodegenerative disorders, thereby improving diagnostic precision [8]. Additionally, SR imaging plays a crucial role in disease diagnosis by enabling clinicians to detect subtle pathological features earlier, which is essential for timely intervention and early screening [52]. Overall, MRI SR not only enhances the visual quality of images but also optimizes subsequent analytical processes and decision-making, facilitating more efficient and accurate clinical diagnosis and treatment while advancing the field of personalized medicine.

In LR MRI images, the loss of edge details is more pronounced. Current methods [17,29,38] largely overlook the utilization of detail and texture information, primarily relying on Transformers for feature modeling. However, transformers emphasize global information, making it challenging to effectively reconstruct satisfactory edge details. Some related works [15,26,40] have started to address these issues by employing CNNs to process high-frequency details while utilizing attention mechanisms to focus on low-frequency information. Nevertheless, these approaches often utilize simple addition or channel concatenation to fuse high-frequency and low-frequency information, which can lead to interference between different frequency components and result in the loss of high-frequency details. Furthermore, this fusion method can create imbalanced feature weights during addition or concatenation, with information from one frequency potentially dominating the fusion result. To address these issues, we propose a HFEB to extract and enhance high-frequency information, while employing a HAB to capture

global features. To effectively fuse high-frequency and low-frequency information, we designed a LWFAB. This block extends the receptive field to capture more features for attention calculations, thereby establishing long-range dependencies and achieving efficient information fusion. Specifically, we represent the output of the HFEB as high-frequency prior queries and the output of the HAB as keys and values, refining and merging the global features produced by the HAB through attention weight calculations. Moreover, within the LWFAB, we utilize a reduction factor to map certain spatial information to the channel dimension through permutation operations, increasing the window size without adding computational overhead, thereby effectively utilizing and fusing high-frequency and low-frequency information.

Additionally, some methods utilize window attention mechanisms for medical image reconstruction. However, during the aggregation of window information, the difficulty of direct information interaction between windows increases the complexity of information integration. Although some approaches introduce overlapping windows to enhance interactions, dense computations of window attention can still result in information redundancy and the introduction of noise when processing large-scale images and complex features, thereby degrading overall performance. To address this issue, we have designed an ASOAB. Each output from the HFEB is multiplied by a learnable residual weight factor, forming a learnable residual branch that generates residual outputs as inputs to the ASOAB, thereby adaptively enhancing features while retaining key information. To eliminate redundant information during the reconstruction process and maintain attention sparsity, we designed an attention branch that utilizes the ReLU activation function. To prevent excessive sparsity, we retained the original Softmax branch and developed a dual-activation function branch. To avoid model instability issues caused by differing learning rates and dynamic adjustments between branches, we apply constraints only to the ReLU branch using a learnable weight factor to adaptively fuse the two branches. This approach effectively reduces redundant information and noise during the reconstruction process, thereby enhancing overall performance.

Finally, we evaluated the FLOPs and inference speed of our algorithm on the BraTS dataset under a $\times 4$ upsampling factor for FLAIR pathological sequences to evaluate its clinical deployability. As shown in Table 2, our algorithm demonstrates 549.05G of FLOPs, a processing speed of 158.6 ms, and a PSNR of 35.64 dB. Although the FLOPs are relatively high, they are comparable to those of other state-of-the-art algorithms, allowing for operation on high-performance medical devices. In terms of processing speed, 158.6 ms is suitable for the vast majority of medical applications, particularly in scenarios requiring high image quality. Our method effectively preserves details, assisting medical personnel in accurately identifying lesions. Our algorithm demonstrates broad applicability, with potential for further optimization through techniques such as model compression and hardware acceleration in the future.

6. Conclusion

This paper proposes a novel brain MRI SR model that effectively utilizes and integrates high-frequency and low-frequency information to extract critical features and reconstruct high-quality SR images. During the image reconstruction process, we designed a High-Frequency Enhancement Block (HFEB) to enhance high-frequency features and developed a Hybrid Attention Block (HAB) to capture global information. Additionally, to better fuse the extracted local and global features, we proposed a Large-Window Fusion Attention Block (LWFAB) that effectively captures additional information while integrating the strengths of CNNs and Transformers. To enhance direct interactions between different window attentions, eliminate redundant information, and adaptively retain critical features, we designed learnable residual weighting branches across multiple Hybrid Fusion Blocks (HFEBs) and proposed an Adaptive Sparse Overlapping Attention Block (ASOAB).

Within the ASOAB, we implemented a dual-branch attention mechanism to enforce sparsity in attention and incorporated a weighting factor to adaptively fuse the two branches. Quantitative and qualitative experimental results validate the superiority of the proposed method over state-of-the-art techniques across multiple datasets and imaging modalities.

However, we found that while the model performs excellently at $\times 2$ and $\times 4$ upsampling factors compared to other state-of-the-art methods, producing satisfactory visual results, producing satisfactory visual results, reconstructing clinically acceptable images in practical applications often requires higher upsampling factors, such as $\times 6$ and $\times 8$. At these higher upsampling factors, as with the other methods, the visual quality of the reconstructed image is significantly reduced. Therefore, we will focus our research on SR techniques at higher upsampling factors in the future.

CRediT authorship contribution statement

Zhiying Yang: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Data curation, Conceptualization. **Hanguang Xiao:** Visualization, Validation, Funding acquisition, Formal analysis. **Xinyi Wang:** Writing – review & editing, Visualization, Supervision. **Feizhong Zhou:** Visualization, Supervision. **Tianhao Deng:** Writing – review & editing, Validation. **Shihong Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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